Therefore the separation distance between the isotopes is $\Delta d = d_{238} - d_{235} = 1.2\,\text{cm}$

(b) The time taken by each isotope to complete one semi-circular path are

$$t_{235} = \frac{\text{magnitude of the displacement}}{\text{velocity}}$$

$$= \frac{97.6\times10^{-2}}{1.00\times10^{5}} = 9.76\times10^{-6}\,s = 9.76\,\mu s$$

$$t_{238} = \frac{\text{magnitude of the displacement}}{\text{velocity}}$$

$$= \frac{98.8\times10^{-2}}{1.00\times10^{5}} = 9.88\times10^{-6}\,s = 9.88\,\mu s$$

Note that even though the difference between mass of two isotopes are very small, this arrangement helps us to convert this small difference into an easily measurable distance of separation. This arrangement is known as mass spectrometer. A mass spectrometer is used in many areas in sciences, especially in medicine, in space science, in geology etc. For example, in medicine, anaesthesiologists use it to measure the respiratory gases and biologist use it to determine the reaction mechanisms in photosynthesis.

3.10.3 Motion of a charged particle under crossed electric and magnetic field (velocity selector)

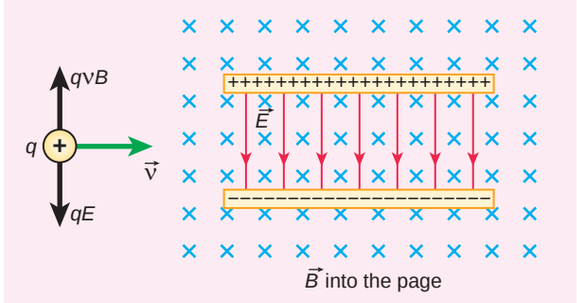


Figure 3.53 Velocity selector

Consider an electric charge q of mass m which enters into a region of uniform magnetic field $\vec{B}$ with velocity $\vec{v}$ such that velocity is not perpendicular to the magnetic field. Then the path of the particle is a helix. The Lorentz force on the charged particle moving in a uniform magnetic field can be balanced by Coulomb force by proper arrangement of electric and magnetic fields.

The Coulomb force acts along the direction of electric field (for a positive charge q) whereas the Lorentz force is perpendicular to the direction of magnetic field. Therefore in order to balance these forces, both electric and magnetic fields must be perpendicular to each other. Such an arrangement of perpendicular electric and magnetic fields are known as cross fields.

For illustration, consider an experimental arrangement as shown in Figure 3.53. In the region of space between parallel plates of a capacitor (which produces uniform electric field), uniform magnetic field is maintained perpendicular to the direction of electric field. Suppose a charged particle enters this space from the left side as shown, the net force on the particle is

$$\vec{F} = q\left(\vec{E} + \vec{v}\times\vec{B}\right)$$

For a positive charge, the electric force on the charge acts in downward direction whereas the Lorentz force acts upwards. When these two forces balance one another, then

$$qE = qv_{\circ}B$$

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$$\Rightarrow v_{\circ} = \frac{E}{B} \qquad (3.61)$$

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